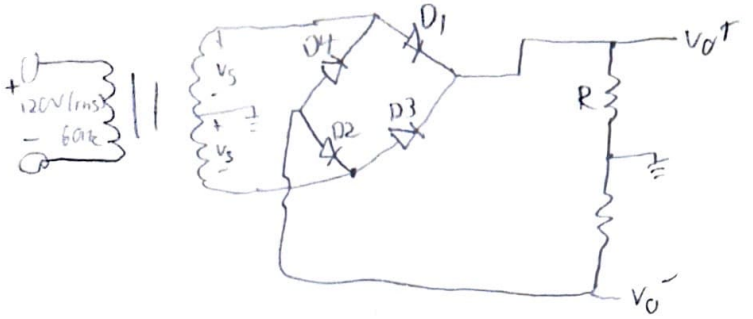
ECEN 115 HW5 Dylan Tromley

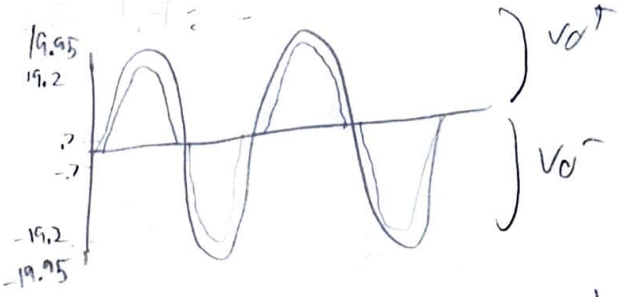
1.



$V_{avg} = 12 = \frac{2x}{\pi} - V_{d0} =$

$12 = \frac{2x}{\pi} - .7$

$\frac{12.7\pi}{2} = x$



$x = 19.95$ B/c it's across all of the winding, we do $19.95 \cdot 2 = 39.9V$

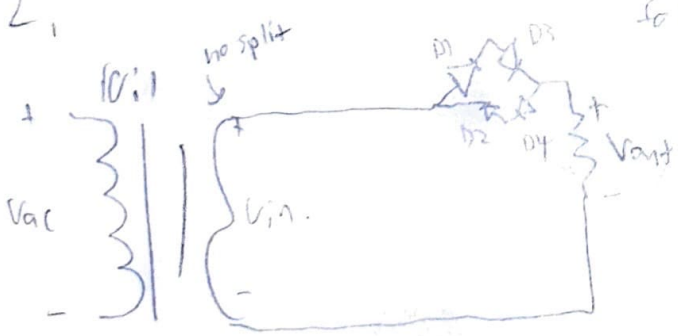
$PIV = V_R = 39.9 - .7 = 39.2V$

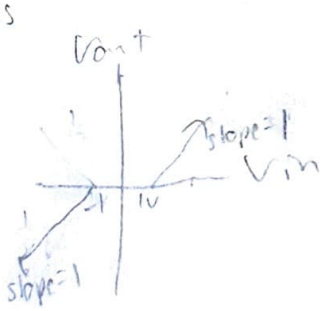
required amplitude = 39.9 V
PIV of each diode = 39.2 V

2. i $120\sin(120\pi t) \rightarrow 12\sin(120\pi t)$

This circuit tries to trick you

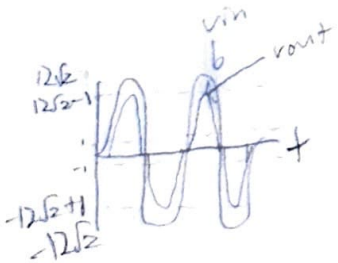
$V_{in} = 2V_{d0}$
$120 \rightarrow 12$ so





$\frac{12}{120} = \frac{1}{10}$ but we only care about vout vs vin not vac

ii



Vout avg just equals 0

D1 = D3 = 3.2mA
D2 = D4 = -3.2mA

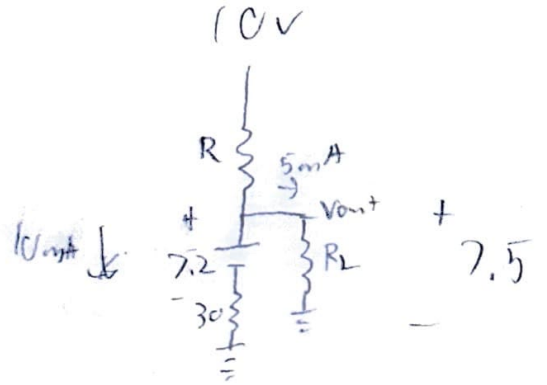
iii He should add a center tap so he actually gets a full wave rectifier.

3.

7.5V zener at 10mA

$r_z = 30$, knee current = .5 mA

10V supply nominal 5 mA to load



$7.5 = V_{30} + 10\times10^{-3} \cdot 30 =$

$7.2 = V_{30}$

$I_z + I_L = 15mA$

$\frac{10-7.5}{R} = 15mA$

$R = 166.67\Omega$

5mA to load

$\frac{7.5}{R_L} = 5\times10^{-3}$

$R_L = 1500\Omega$

$1500 \parallel 30 = 29.4117$

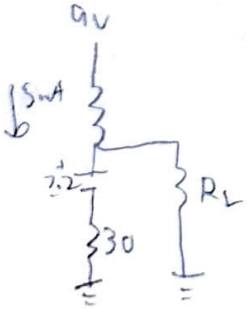
$10\left(\frac{29.4117}{166.67+29.4117}\right) = 1.5$

$11\left(\frac{29.4117}{166.67+29.4117}\right) = 1.65 - 1.5 = +.15$

$9\left(\frac{29.4117}{166.67+29.4117}\right) = 1.35 - 1.5 = -.15$

w/ 10% high, 7.65V

10% low, 7.35V



$5mA(30) = 15mV$

$15mV + 7.2 = 7.215$

$\frac{9-7.215}{166.67} = 10.71mA$

$10.71 - .5 = 10.21$

max load current =

10.21mA

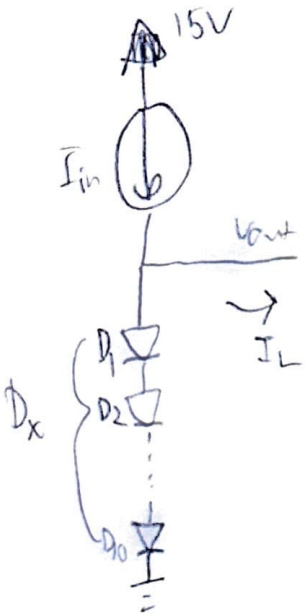
$7.2 + .015 = 7.215$

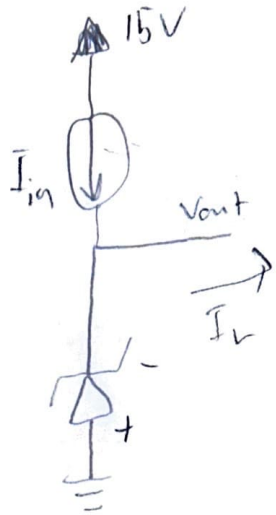
$V_L = 7.215V$

when load is removed and 10% high

$\left(\frac{11-7.2}{166.67+30}\right)(30) + 7.2 = 7.78V$

11.





6V at 5mA

$r_2 = 200$

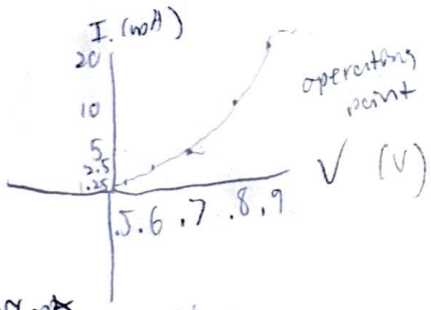
$I_{2A} = .5\,mA$

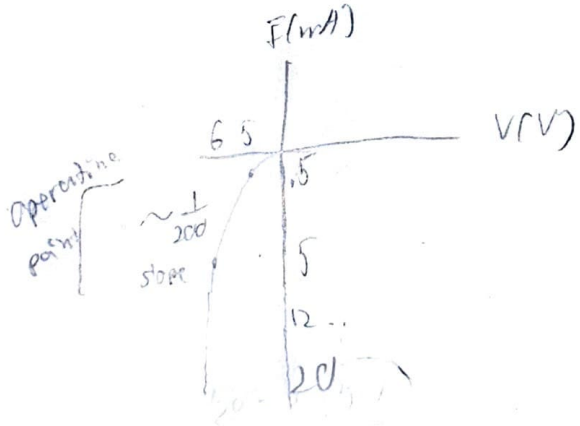
$6 = V_{20} + 200 * 5mA$

$V_{20} = 5$

D_2 |

D_x





ii at 20mA

$10 * .9 = \boxed{9V}$

ii at 20mA

$\boxed{9V}$

20mA - 8mA = 12mA

12mA → .82

$.82 * 10 = 8.2$

$\frac{8.2}{9}(100) = \boxed{8.89\% \text{ change}}$

$(.02 - .005) = \frac{1}{200}(x - 6)$

$x = 9$

iii

$(.02 - .005) = \frac{1}{200}(x - 6)$

$x = 7.4$

$\frac{7.4}{9}(100) = \boxed{17.78\% \text{ change}}$

20mA to 5mA will vary it 20-15

7-9V

$\left(\frac{9-7}{9}\right)100 = 22.2\%$ ← lower load regulation %

20mA to 5mA will vary

6-9V

$\left(\frac{9-6}{9}\right)(100) = 33.3\%$

iv. Figure 2a (the stark) is a better regulator b/c it's load reg. % is smaller. It has less voltage variance for a load.

V.

5mA vs 10mA vs 40mA vs 20mA vs 80mA

7V 8V 10V 9V 11V

exponential gap

8 – 9 – 10

Picking 40 mA is better b/c a greater variance in A at this point will yield a smaller change in voltage

+20 40 +40 80

20 → 40 → 80

+1V +1V

5mA vs 10mA vs 40mA vs 20mA vs 80mA

6V 7V 13V 9V 21V

linear gap

7 – 9 —— 13

Choosing 10mA or 40mA doesn't matter heart when b/c the regulation will remain linear regardless.